

PHY180 Final Lab: Analyzing Period, Amplitude, Length, and Q Factor in a Homemade Pendulum

Michelle Liu

Professor Brian Wilson

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I. INTRODUCTION

The homemade simple pendulum acts as a damped oscillator. In this lab, the pendulum was created with a key tied to the end of a length of floss. Pendulum motion was captured on video at 60 fps using an iPhone 13, to then be mapped and analyzed using Tracker software (Brown, 2025a). The camera was positioned level to the pendulum's pivot point to ensure stable and optimal recording of motion. The pendulum's damped harmonic motion can be modeled with Equation 1:

$$\theta(t) = \theta_0 e^{-t/\tau} \cos\left(2\pi\frac{t}{T} + \phi_0\right) \quad (1)$$

where θ is the angle of release, t is the time, τ is the time constant for decay, and T is the period. In Lab 1, we verified that Equation 1 is viable, based on the period vs. amplitude relationship found. The period vs. angle relationship is modeled with Equation 2:

$$T(\theta) = T_0(1 + B\theta + C\theta^2) \quad (2)$$

where B and C are constants, with $B = 0.0002 \pm 0.002$ and $C = 0.053 \pm 0.003$. The B value describes symmetry, where $B = 0$ indicates a symmetric pendulum. We determine B to be experimentally 0 because the B value is less than the uncertainty. Therefore, the pendulum is symmetric. The C value is representative of the period's dependence on amplitude. C was determined to be inconsistent with 0 at amplitudes greater than $\pi/6$ because it was more than 2 times its uncertainty. At angles smaller than $\pi/6$, C is within 2 times its uncertainty, $C = 0.09 \pm 0.05$, thus the period is only independent of amplitude at small angles. In Lab 1, we also determined the Q factor, which is the quality factor describing the pendulum dampening. Larger Q factors correspond to less energy loss and less amplitude decay over time. For the Tracker method of finding Q , we use Equation 3:

$$Q = \pi \frac{\tau}{T} \quad (3)$$

Q Factor was found using 2 methods. First, amplitude was plotted against time, using Tracker software (Brown, 2025a). The value of τ was found by fitting the data to an exponential function, where τ is the time constant for amplitude decay. Equation 3 was then used

to find $Q1 = 146 \pm 2$. In the second method, number of pendulum oscillations was counted until the amplitude decreased to $e^\pi/2$ to get $Q2/2$. $Q2$ was then found to be 131 ± 2 . The method of using Tracker to find Q factor was more effective given that this method had lower percentage uncertainty of 0.71% whereas the counting method had uncertainty 1.54%. The Tracker method of finding Q was thus chosen for Lab 2.

In Lab 2, period-length and Q factor-length relationships were analyzed. The period and length had a square root relationship. The graph of period versus length was fitted to a power-law function, described by Equation 4:

$$T = kL^n \quad (4)$$

where k and n are constants. The predicted constants were $k = 2$ and $n = 0.5$. The actual data showed $k = 1.97 \pm 0.05$ and $n = 0.46 \pm 0.02$. The k and n values are consistent with the predictions since they are within 2 standard deviations of 2 and 0.5, respectively. The power-law function was then plotted on a logarithmic scale. The log-log graph explicitly showed a linear relationship between the log scale values of the power function and the period.

When plotting Q and length, they were found to have a linear relationship, where Q decreased linearly with length. The linear fit was determined after attempting to plot the Q vs. length relationship with a power series fit, exponential fit, and linear fit. When expressed as percentages, the uncertainties of parameters a , b , and c of the power series fit were 3.80%, 7.97%, and 28.66% respectively. For the parameters a , and b of the exponential fit, percentage uncertainties were 2.55% and 85,000,000% respectively. For the linear fit $a = 1930 \pm 50$ and $b = 950 \pm 20$, with percentage uncertainties of 2.67% and 1.60%, respectively. Comparing the percentage uncertainties, the linear fit had significantly smaller uncertainty, therefore it was the best fit.

II. METHODS

A. EXPERIMENTAL SETUP



Fig. 1. Image of pendulum experimental setup, iPhone 13 placed 0.68 ± 0.005 m away from pivot point of pendulum.

To create the pendulum shown in Figure 1, a key was tied to a piece of floss. The other end of the floss was secured in a key ring, which was taped to the wall. A protractor was taped level to the pendulum's pivot point, a distance away to avoid contact with the pendulum. The iPhone 13 used for video capture was placed 0.68 ± 0.005 m. These design decisions were justified given that the pendulum was free to oscillate without risk of interference from a wall or object, with a metal key ring used to minimize friction at the pivot point. The camera was positioned level to the pendulum's pivot point to avoid parallax errors.

B. UNCERTAINTIES

In the overall experiment, type A uncertainties included: inconsistencies in perceiving the point of a full oscillation, variations in the point of the key being tracked, and fluctuating air currents due to the vent next to the pendulum (see Fig. 1). Type B uncertainties included: protractor angle increments, video frames per second, ruler increments, air resistance, friction at the pendulum's pivot point, and parallax errors. Many of the uncertainties were found to be experimentally insignificant (such as air current, air resistance, and friction) after performing numerous trials for data collection.

C. PROCEDURES

Part 1: Period vs. Amplitude

To find the period corresponding to each angle, firstly a video of the pendulum's motion was captured on an iPhone 13 at 60 fps. The video was imported into Tracker and the frame at which the pendulum reached the top of its swing was recorded. The number of frames for the oscillation starting and ending at the top of the swing was determined and multiplied by 1 sec/60, the frame rate. The average of 5 trials were taken, to reduce uncertainties. The type B uncertainty for the period was determined to be half a frame, or 1 sec/120. Python program fit black box.py (Wilson, 2025c) was used to create Period vs. Amplitude graph.

This method of finding period was justified because it eliminated uncertainties of reaction time, being able to analyze the video frame by frame. The period data was not affected by amplitude decay either because the period was measured for the accurate, undecayed amplitude using video.

Part 2: Q Factor

For the first method of finding Q, using Tracker, the pendulum's motion was recorded using the iPhone. The Physlets Tracker application analyzed the video and plotted the position of the pendulum over the first 100 seconds, at 60 fps, after being released from rest. Python program fit black box.py (Wilson, 2025c) was used to create Amplitude vs. Time graph using peaks of Tracker's sinusoidal graph. The video decisions made were justified because 100s is long enough to gather data for > 150 peaks, allowing for an accurate fit to find τ . Recording at 60 fps rather than 30 also reduces type B uncertainty due to the video frames. For the second method of finding Q, counting oscillations, the number of oscillations were counted until angle decreased to 20%, then multiplied by 2.

Part 3: Period vs. Length

To find the period corresponding to each length, the same method was used as finding periods corresponding to amplitudes, for a constant amplitude of $\pi/6$. The amplitude was chosen based on results from Lab 1 showing that an assumption of consistent period for small angles ($\leq \pi/6$) is valid. Thus, amplitude decay is

minimized for this data.

Part 4: Q Factor vs. Length To find Q Factor corresponding to each length, the Tracker method was used, as chosen by the lower uncertainty when expressed as a percentage (0.71%), compared to the counting oscillations method (1.54%)

III. RESULTS & ANALYSIS

A. PERIOD VS. AMPLITUDE

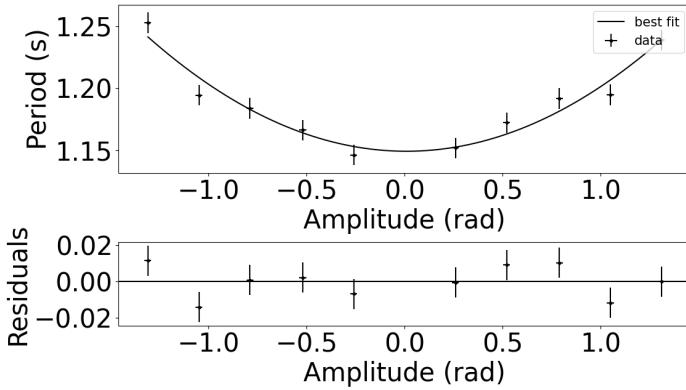


Fig. 2. Period (s) vs. Amplitude (rad) graph for 10 angles from $-5\pi/12$ rad to $5\pi/12$ rad at intervals of $\pi/12$ with quadratic best fit curve, generated using Python script. The fit of data to the power series (quadratic, $y = T_0(1 + (0.00020.002)\theta + (0.0530.003)\theta^2)$) shows variance in period with changes in amplitude. Amplitude uncertainties are too small to see on graph, uncertainty = 0.02 rad

Figure 2 shows the period's dependence on the pendulum's amplitude. It also displays the fit of a power series. The quadratic is a valid fit given that the residuals graph (Fig. 2) shows 60% of data points within 1 standard deviation. The power series is described by Equation 2, with values of B and C from the Python script. While the prediction for B and C

values was $B = C = 0$, actual values of B and C were $B = 0.0002 \pm 0.002$ and $C = 0.053 \pm 0.003$. Since the B value is smaller than its uncertainty, it is experimentally 0. Given that $B = 0$, the pendulum is thus symmetric.

Since the C value is more than 2 larger than its uncertainty, it is not consistent with 0. With a C value unequal to 0, the pendulum's period cannot be independent of amplitude, as the power series fit is not a

flat line with zero slope. The prediction of the period's independence is therefore refuted. When restricting amplitude to smaller angles and finding the associated C values, angles from $-\pi/6$ to $\pi/6$ are consistent with zero. At $\pm\pi/6$, $C_{\pi/6}$, where C is less than 2 times larger than the uncertainty. In that range of amplitudes, C is small enough to be ignored and the prediction of a zero slope line of best fit is valid.

B. Q FACTOR

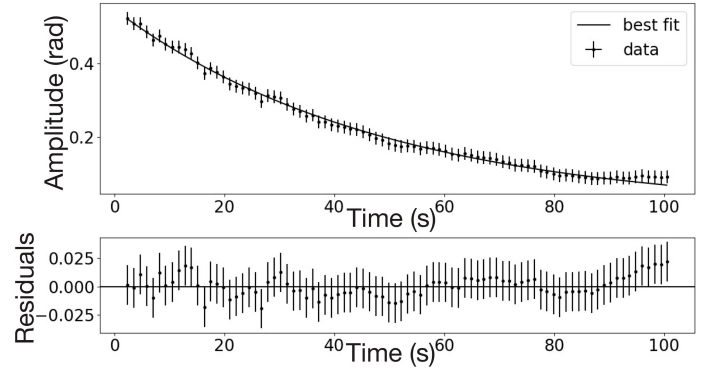


Fig. 3. Amplitude (rad) vs. Time (s) graph of peaks of pendulum oscillations with exponential best fit curve, generated using Python script. The fit of data to the function shows the exponential decay of the amplitude due to friction, $\tau = 54.8 \pm 0.8$. Uncertainties for amplitude and time are 0.02 and 0.03, respectively. Time uncertainties are too small to be visible, given the length of time graphed (0.008 s uncertainty against 100 s on graph).

Figure 3 shows the decaying amplitude as it relates to time. It is fitted with an exponential function, a valid fit given that the residuals graph shows 90% of data within 1 standard deviation. The equation of the exponential function is described by Equation 1. The phase constant $\phi = 0$ given that the time started when the pendulum was released. From data for the Period vs. Amplitude graph (Fig. 2), $T = 1.172 \pm 0.007$.

From the Python scripts, τ was determined to be $\tau = 54.8 \pm 0.8$. The value of τ can be used for the first method of finding Q (Q1, method using Tracker). The Q1 factor for the pendulum is determined by Equation 3 (Wilson, 2025b) below, where $T = 1.172 \pm 0.008$.

$$Q1 = \pi \frac{\tau}{T} \quad (5)$$

$$Q1 = 146 \pm 2$$

The uncertainty of Q was the uncertainty of τ expressed as a percentage, the larger of the percentage uncertainties between τ and period. Percentage $(\text{uncertainty } \tau)/\tau = 1.54\%$, where the τ uncertainty was determined with Python script. Percentage $(\text{uncertainty period})/\text{period} = 0.71\%$ (see Fig. 1 for period and associated uncertainty value).

For the second method of finding Q (Q_2 , method counting oscillations), the number of pendulum oscillations was counted until the amplitude decreased to $e^\pi/2$ to get $Q/2$. The counted $Q/2$ value had an uncertainty of 1 to account for the experimenter's potential errors when counting integers. Using this method, $Q_2 = 131 \pm 2$. To determine if the two methods of finding Q agree, the absolute difference of Q values can be determined with Equation 4 below.

$$\begin{aligned} \Delta Q &= |Q_1 - Q_2|T \\ &= |146 - 131| \\ &= 15 \end{aligned} \quad (6)$$

Since the absolute difference is greater than 5 times the larger uncertainty, the Q values do not agree, therefore the two methods of finding Q do not align. Given the lower uncertainty of Q_1 (when expressed as a percentage), the Tracker method is more valid for determining Q value.

C. PERIOD VS. LENGTH

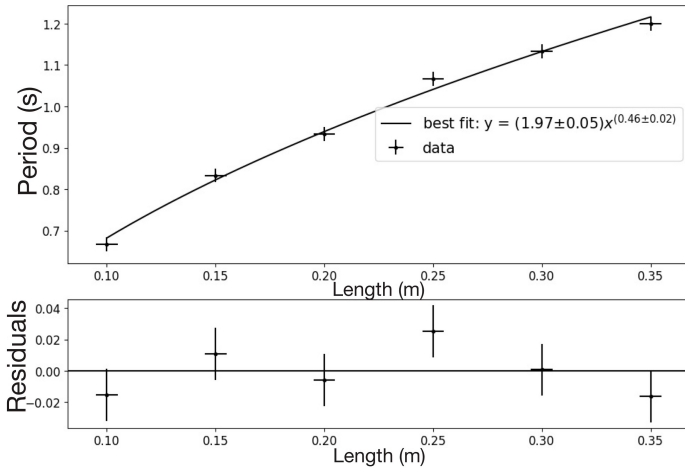


Fig. 4. Period (s) vs. Length (m) graph showing square-root relationship of data, for 6 lengths. Power fit function is $y = (1.07 \pm 0.05)x^{(0.46 \pm 0.02)}$ and shows the period's dependence on length.

Figure 4 shows that the period of the pendulum is dependent on length. It displays period and length fitted to a power function (square root relationship). The square root fit is valid given that around 67% of data points are within 1 standard deviation as shown in the residuals plot (Fig. 4). The graph is modeled by Equation 4, where the prediction for k and n values was $k = 2$ and $n = 0.5$. (Wilson, 2025b). Actual values were found to be $k = 1.97 \pm 0.05$ and $n = 0.46 \pm 0.02$. The predicted and experimental k values have an absolute difference of 0.03. This difference is within 1 standard deviation of the experimental k , since $0.03 < 0.05$. The predicted and experimental n values have an absolute difference of 0.04. This difference is within 2 standard deviations of the experimental n , since $0.04 = 0.02 \cdot 2$. Given that the k value is within 1 standard deviation of the prediction and the n value is within 2, the data is consistent with the prediction and the expression $T = 2\sqrt{L}$ is valid.

D. LOG PERIOD VS. LENGTH

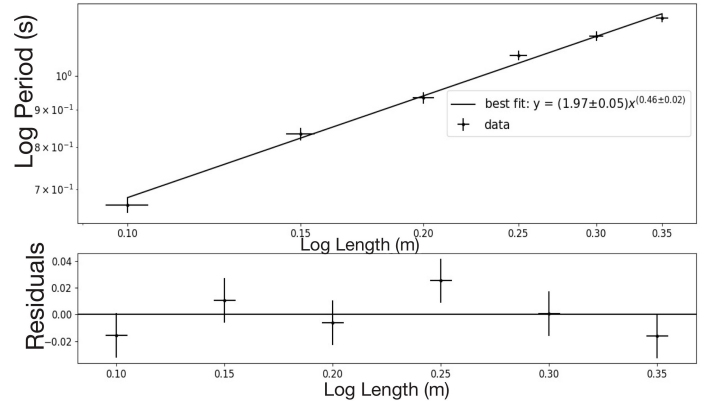


Fig. 5. Log Period vs. Log Length graph showing linear relationship of data on a log scale. Linear fit function has $y = (0.46 \pm 0.02)x + \log(1.97 \pm 0.05)$ and explicitly shows period's dependence on length.

Figure 5 shows the period vs. length relationship plotted on a logarithmic scale to linearize the graph. The linear relationship between log period and log length provides explicit evidence that the period is dependent on length with a power relationship.

E. Q VS. LENGTH

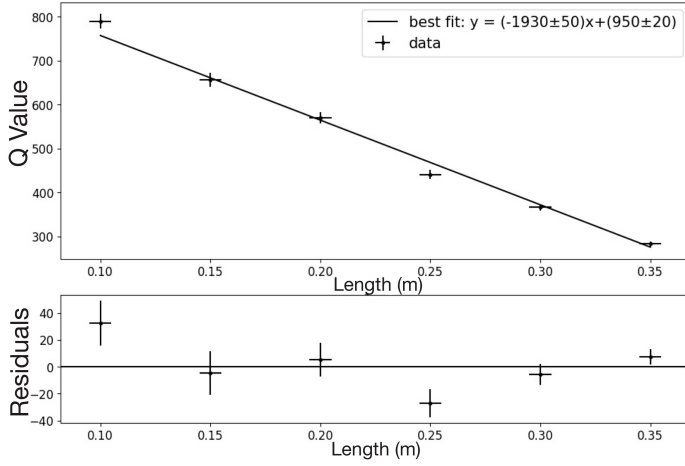


Fig. 6. Q Factor vs. Length (m) graph showing linear relationship, with Q Factor decreasing with length. Linear fit function has $y = (-1930 \pm 50)x + (950 \pm 20)$. Q Factor uncertainties vary based on τ value uncertainties found through Python code.

Figure 6 shows that Q factor and length have a linear relationship, with Q decreasing linearly with length. Given that around 67% of data is within 1 standard deviation, as shown by the residuals plot, this relationship is valid. Q and length were attempted to be plotted with other relationships, including a power series and exponential fit. The power series fit had $a = 1070 \pm 3.80\%$, $b = 2.8 \pm 7.97\%$, $c = 2.0 \pm 28.66\%$. The exponential fit had $a = 400 \pm 2.55\%$, $b = 15,000,000 \pm 85,000,000\%$. The linear fit had $a = 1930 \pm 2.67\%$, $b = 950 \pm 1.60\%$. The linear fit has the lowest percentage uncertainties, therefore, we can conclude that Q factor and length have a linear relationship where Q factor decreases with length.

Values of Q were found using the Tracker method, because it yielded lower uncertainties than the counting method from Lab 1, when expressed as a percentage.

IV. DISCUSSION & IMPROVEMENTS OF UNCERTAINTIES

The data for the Period vs. Amplitude graph (Fig. 2) included type A and B uncertainties. The most significant was measurement of amplitude, which was an uncertainty of 1 degree given the protractor's increments of 0.02 rad. When expressed as a percentage of the amplitude values, the uncertainty is around

2% which is significantly larger than that of the video's frame rate uncertainty of 0.69% for period values. Although they are both type B uncertainties, uncertainty in amplitude measurement encompasses uncertainties from perception of a full oscillation. In Tracker, it can be difficult to see the protractor's increments in relation to the pendulum's string, causing error in registering an oscillation. As outlined in the procedure, taking multiple trials does help reduce this error.

To improve the uncertainty measuring amplitude, angle measurements can be drawn onto a larger surface, like a piece of paper when collecting amplitude data. This larger surface can be placed behind the pendulum, instead of relying on a protractor with small, potentially unclear increments.

The data for the Amplitude vs. Time graph (see Figure 3) used type B uncertainties, as they were most plausible. The uncertainty for amplitude was once again the protractor increments of ± 0.02 rad. The uncertainty for time was the length of 1 frame = ± 0.008 s, when using video and Tracker (Brown, 2025a). The uncertainty when counting oscillations was ± 2 oscillations—where the error of counting Q/2 was 1 full oscillation and it was propagated to Q.

To improve uncertainties when using Tracker, a specific tracking point (such as a point drawn onto the key in Fig. 1) can be depicted on the mass to ensure accurate position tracking. Ensuring consistent tape placement and experimental set up will help reduce changes in energy loss and Q factor.

The Length vs. Period graph had similar uncertainties to that of the Period vs. Amplitude graph, but also included type B uncertainties from measuring the lengths of floss. In addition to the uncertainty of half a cm (0.005 m) from the ruler, there was some slight changes in string length due to the way it was tied to the key at the bottom and secured to the wall at the top. Thus, there were errors in the string length and more uncertainties.

The Q Factor vs. Length graph had similar uncertainties to that of the uncertainties when finding

Q on its own. Additional errors could have been introduced when changing the length of the string. At some lengths, more tape was added to secure the string, potentially causing more friction with the pendulum's swing. With more energy lost to friction, there would be increased amplitude decay and the Q Factor would decrease, causing potential errors and increasing uncertainties.

To improve uncertainties with length and pivot point friction, new pieces of string can be cut for each length, instead of simply tying up the same length of string. Each string could then be marked at the appropriate length, with the marking placed at the pivot point, such that the actual string length of the pendulum is accurate. With consistent tape placement, energy lost to pivot point friction can thus be reduced.

V. CONCLUSION

Upon analysing pendulum data for the Period vs. Amplitude (Fig. 2), Amplitude vs. Time (Fig. 3), Period vs. Length (Fig. 4), and Q Factor vs. Length (Fig 6), the validity of initial predictions can be determined.

The quadratic relationship between Period vs. Amplitude (Fig. 2, Eq. 2), shows that period is dependent on amplitude, where $B = 0.0002 \pm 0.002$ and $C = 0.053 \pm 0.003$. Since B is experimentally 0, the pendulum is symmetric, but C is not consistent with 0, therefore refuting the initial prediction that period is independent of amplitude. At angles between $-\pi/6$ and $\pi/6$, it is valid to approximate that period is independent from amplitude, because $C_{\pi/6} = 0.09 \pm 0.05$, where the value of $C_{\pi/6}$ is less than twice the uncertainty

The Tracker method for finding Q resulted in $Q1 = 146 \pm 2$ and the counting method resulted in $Q2 = 131 \pm 2$. The two methods have an absolute difference of 15, which is greater than 5 times the uncertainty, since $5 \cdot 2 = 10$. The measurement methods do not agree with each other because of the difference in the Q values. Future data collection for Q value will be impacted by choosing to use the Tracker method, given the lower uncertainty of Q factor when expressed as a percentage.

In Lab 2, the period was found to be dependent on length, with a square root relationship between length and period. The experimentally determined k and n values were: $k = 1.97 \pm 0.05$ and $n = 0.46 \pm 0.02$. The predicted value $k = 2$ has an absolute difference of 0.03 with the experimental k value, which is within the uncertainty of experimental k, given $0.03 < 0.05$. Thus, the k values are consistent and the prediction is valid. The predicted value $n = 0.5$ has an absolute difference of 0.04 with the experimental n value, which is within 2 times the uncertainty of experimental n, given $0.04 = 0.02 \cdot 2$. Thus, the n values are consistent and the prediction is valid. The expression $T = 2\sqrt{L}$ is accordingly valid.

The relationship between Q factor and length was determined to be a linear decreasing relationship. Upon trying a power series, exponential, and linear fit, we found that the parameters of the power series a, b, c had percentage uncertainties 3.80%, 7.97%, 28.66%, respectively, the parameters of the exponential a, b had percentage uncertainties 2.55%, 85,000,000%, and the parameters of the linear a, b had percentage uncertainties 2.67%, 1.60%. The fit of Q factor to a linear function was most valid given the lowest and most reasonable uncertainties when expressed as a percentage, as well as the valid residuals discussed.

From the discussion of uncertainties, we can understand improvements for future data collection. can be improved to reduce the most significant uncertainties: amplitude measurement for period-amplitude relationships, data collection in Tracker, and experimental set up for Q values. When collecting data for amplitude, angle values can be drawn onto a larger surface for clarity of analysis. To reduce uncertainties in collecting Tracker position data, a tracking point can be marked on the mass. Cutting new pieces of string for each length and ensuring consistent tape placement will help reduce changes in energy loss and Q factor. These improvements will contribute to lower uncertainties in data collection.

APPENDIX

REFERENCES

- Brown, Douglas, “Tracker”, URL: <https://physlets.org/tracker/>, 2025a
- Wilson, Brian, “PHY180 Lab Project”. Downloaded from Quercus, 2025b
- Wilson, Brian “fit_black_box.py”. Downloaded from Quercus, 2025c